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High-Frequency Seismogram Envelope Inversion Using a Multiple Nonisotropic Scattering Model: Application to Aftershocks of the 2008 Wells Earthquake

by Yueling Jing, Yuehua Zeng, and Gao Lin

Abstract In this paper, we use the Sato (1995) nonisotropic scattering formulation to describe an intensity spectral density with a spherical impulse wave radiated from the source. A characteristic source time is introduced to define the initial impulse width. We apply the modified multiple nonisotropic scattering model to analyze velocity seismogram envelopes of the 2008 Wells earthquake in Nevada. A series of scattering patterns is generated by a nonisotropic scattering pattern function. Higher-order spherical functions are included in the expansion of the scattering pattern function so that stronger nonisotropic scattering patterns can be represented. A variety of synthesized envelopes, from backward-dominated, forward-dominated, and isotropic scattering processes, are investigated. By comparing synthesized envelopes with observed data, we found that the backscattering pattern fails to predict the observations of high-frequency *S* waves, and envelopes from a forward-dominated scattering pattern provide the best fit to the data. Based on these results, we offer a new interpretation for the origination of high-frequency *S*-coda waves. The broadening effect of the envelope is successfully explained by the process being dominated by forward scattering. For all the aftershocks, the frequency dependence of intrinsic attenuation and scattering attenuation is discussed for frequency bands of 1–2, 2–4, 4–8, and 8–16 Hz. Both types of attenuation decrease when frequency increases, and the intrinsic attenuation dominates over scattering attenuation at higher frequencies.

Introduction

In order to investigate the inhomogeneity of the Earth from the observations of the seismic coda wave, many scattering models have been developed by seismologists. By assuming the earth as a random scattering medium, Aki (1969), Aki and Chouet (1975), and Sato (1977) have studied the seismic coda as single backward scattering waves. Since then, the single backward scattering model has been widely used to interpret observations and to calculate coda-wave attenuation. However, as lapse time increases, double-, triple-, and higher-order multiple scattering contributions become more dominant than the single scattering process. Therefore, a model is needed to include the effect of multiple scattering. Kopnichev (1977) and Gao *et al.* (1983) studied the multiple isotropic scattering process by adding higher-order scattered wave contributions; however, the total energy is not conserved in their models. Wu (1984) first introduced stationary radiative transfer theory as a model for the multiple scattering problems. This theory can be used to separate the effect of scattering and intrinsic attenuation uniquely as demonstrated by Wu and Aki (1988) for the Hindu Kush region. At the same time, from the analysis of 2D finite-difference simulation of wave propagation in inhomogeneous media, Frankel and Clayton (1986) observed the excitation of coda waves. Based on their numerical simulation and assuming a spatially uniform distribution of coda energy density, Frankel and Wennerberg (1987) proposed an energy flux model to calculate the spatiotemporal distribution of coda energy density. Zeng *et al.* (1991) extended the stationary radiative transfer theory studied by Wu (1984) to the time-dependent case in 3D space and formulated a fundamental scattered wave energy equation.

Radiative transfer theory models the transfer of energy in the media; this energy approach is associated with particular phenomenological concepts developed in studying electromagnetic radiative transfer in absorbing, emitting, and scattering media, in which it can take into account both forward and backward scattering processes. However, for mathematical simplicity, multiple isotropic scattering is assumed in most seismic data analyses (Padhy *et al.*, 2007; Lee *et al.*, 2010); with this assumption, the theory only partly models the development of the coda, and the envelope broadening cannot be explained by isotropic scattering process. To

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solve this problem, in the past few decades, many advances have been made with multiple nonisotropic scattering models. Sato (1994, 1995) first formulated the multiple nonisotropic scattering process in the framework of radiative transfer theory. Gusev and Abubakirov (1996) pointed out that by using nonisotropic scattering coefficients (which take into account the strong forward scattering), radiative transfer theory can also simulate the broadening of envelopes, but their numerical results show an unrealistically fast decay in the early coda wave after the direct wave group. Hoshiba (1995) studied the effect of nonisotropic scattering using a Monte Carlo simulation in western Japan, although their work focused mostly on the spatial energy distribution of seismic coda waves. Wegler *et al.* (2006) made a numerical comparison between five different theories for the computation of theoretical seismogram envelope, which include Markov approximation, diffusion approximation, radiative transfer theory with isotropic scattering coefficients, radiative transfer theory with Born scattering coefficients, and the finite-difference method. Finally, they proposed that radiative transfer theory with Born scattering coefficients can model full seismogram envelopes from the first onset to the multiple scattered late coda. However, there is no further data analysis in their comparisons.

Here, in the framework of radiative transfer theory and using the formulation of nonisotropic scattering process (Sato, 1995), an intensity spectral density formulation of the initial impulse wave was incorporated into Sato's original energy density integral equation. Based on the modified energy density formulation, we will give a new interpretation of high-frequency coda-wave excitation from the inversion of the *S*-wave velocity envelope of the 2008 Wells, Nevada, earthquake. In the inversion, a recursive formula is proposed to improve the calculation of higher-order Green's functions in the wavenumber–Laplace domain. Then, total scattering coefficient and characteristic source time, which is used to define the initial energy density impulse duration, are obtained by using a grid-search method over the model space. An intrinsic attenuation factor is estimated by the least-square method. Based on the new scattering mechanism of coda-wave excitation, envelope broadening is reproduced in the synthesized envelope. Finally, for different aftershocks, intrinsic attenuation and scattering attenuation are estimated separately for frequency bands 1–2, 2–4, 4–8, and 8–16 Hz in the Wells region of Nevada.

Theory Background

In this section, some known results will be briefly presented to form the basis of our model. As with multiple nonisotropic scattering theory (Sato, 1995; Sato and Fehler, 1998), randomly inhomogeneous media will be modeled as homogeneous background media that are filled with homogeneously and randomly distributed point-like scatterers. Background media is characterized by the average propagation velocity V_0. The wavefield, which originates from the interaction of seismic waves and these randomly distributed scatterers, is incoherent. Therefore, the energy density of the scattering wave can be added, and the nonisotropic scattering process can be described in the framework of radiative transfer theory. According to Sato's nonisotropic scattering formulation, the directional distribution of the scattered energy flux around a scatterer is assumed to be axially symmetric and nonisotropic. As shown in Figure 1, this scattering process is characterized by a directional scattering coefficient $g(\psi) = g_0\zeta(\psi)$, in which g_0 is the total scattering coefficient, $\zeta(\psi)$ is the scattering pattern function, and ψ is the scattering angle measured from the direction of the incident waves. The directional distribution of energy density at $\mathbf{r}$ in direction $\mathbf{q}$ at time t is defined by $f(\mathbf{r}, t; \mathbf{q})$. When a total energy W is spherically radiated instantaneously from the origin at $t = 0$, the $f(\mathbf{r}, t; \mathbf{q})$ can be formulated in spherical coordinates as (Sato, 1995)

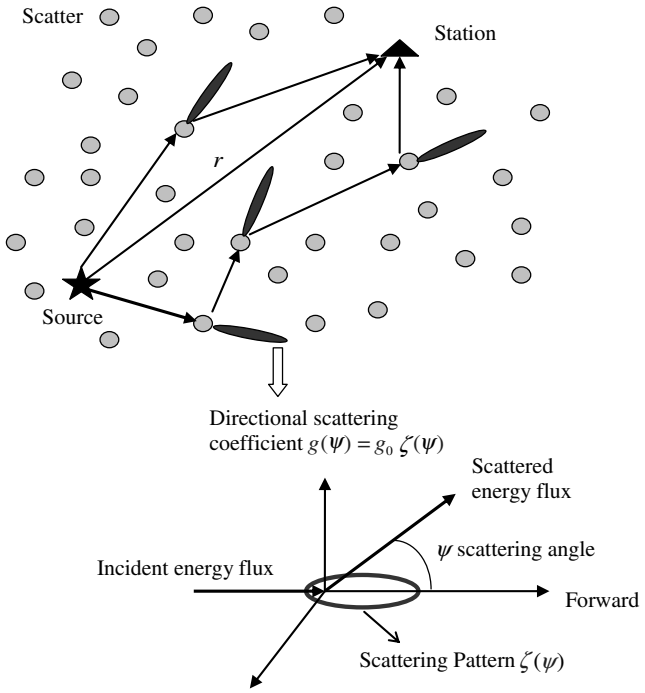


Figure 1. A schematic plot of the multiple nonisotropic scattering process.

$$f(\mathbf{r},t;\mathbf{q}) = WG(\mathbf{r},t;\mathbf{q}) + V_0 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \oint G(\mathbf{r}'', t-t';\mathbf{q}) \times g(\psi) f(\mathbf{r}', t';\mathbf{q}')\, d\mathbf{r}'\, dt'\, d\Omega(\mathbf{q}'). \quad (1)$$

The first term in equation (1) is the contribution due to direct propagation between the source and receiver. $\mathbf{q} = (1, \theta_q, \varphi_q)$ is a unit vector in spherical coordinates, which is used to denote the direction of energy flux at receiver, and $G(\mathbf{r}, t; \mathbf{q})$ is a directional Green's function, which expresses the distribution of energy density in direction $\mathbf{q}$ due to direct propagation from a spherical impulsive radiation of unit energy located at the origin at $t = 0$. Its representation is

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$$G(\mathbf{r},t;\mathbf{q}) = \frac{e^{-(g_0V_0+b)t}}{4\pi V_0 r^2}\delta(t-r/V_0)H(t)\delta_\Omega(\mathbf{q};\mathbf{r}), \quad (2)$$

in which b is the intrinsic attenuation factor to account for the intrinsic attenuation per time in the media, g_0V_0 is for scattering attenuation, δ_Ω is the delta function for solid angles (see the definition in equation A1).

The second term in equation (1) is a convolution integral characterizing the energy flux in direction $\mathbf{q}'$ that hits the last scatterer, and the scattered energy flux proportional to $g(\psi)$ is excited in direction $\mathbf{r}''$. The directional Green's function in the convolution integral is used to describe the propagation energy from the last scatterer to the receiver. The details of geometry of the nonisotropic scattering process are illustrated in Figure 2. The integral of $f(\mathbf{r},t;\mathbf{q})$ over a solid angle with respect to $\mathbf{q}$ gives the total energy density:

$$E(\mathbf{r},t) = \oint f(\mathbf{r},t;\mathbf{q})d\Omega(\mathbf{q}). \quad (3)$$

For spherically outgoing waves in 3D random media, in order to model the initial energy density (or initial mean square [MS] envelope) in frequency domain $I(\omega)$ at the source radiation, we use the analytical functional form given by equation (4), in which parameter t_M is a constant showing the duration of the initial envelope. The initial energy density is incorporated into the Sato's nonisotropic scattering theory. We note that this functional form is borrowed from the energy density through random media, with a Gaussian autocorrelation function based on the Markov approximation by Shishov (1974) and Saito *et al.* (2002) for the case that t_M is proportional to the square of travel distance. In our paper, the quantity t_M is independent of the medium heterogeneity. $I(\omega)$ is reproduced as

$$I(\omega) = \frac{2e^{i\pi/4}\sqrt{t_M\omega}}{\sin\{2e^{i\pi/4}\sqrt{t_M\omega}\}}. \quad (4)$$

Meanwhile, normalized lapse time, hypocentral distance, and energy density are used, so that the nonisotropic scattering theory of Sato can be applied directly. These normalized parameters are as follows:

$$\tau = V_0g_0t, \qquad \boldsymbol{\rho} = g_0\mathbf{r}, \qquad \bar{f} = \frac{f}{Wg_0^3},$$
$$\bar{E} = \frac{E}{Wg_0^3}, \qquad \bar{b} = \frac{b}{V_0g_0}. \quad (5)$$

Then, the modified energy density can be obtained by the convolution of $I(\omega)$ and $\bar{E}_{\text{Sato}}(\boldsymbol{\rho},\omega)$, which is the representation of the normalized energy density of Sato (1995)

$$\bar{E}(\boldsymbol{\rho},\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega e^{-i\omega\tau}\{\bar{E}_{\text{Sato}}(\boldsymbol{\rho},\omega) * I(\omega)\}, \quad (6)$$

in which

$$\bar{E}_{\text{Sato}}(\boldsymbol{\rho},\omega) = \int_{-\infty}^{\infty} d\tau e^{i\omega\tau}\bar{E}_{\text{Sato}}(\boldsymbol{\rho},\tau) \quad (7)$$

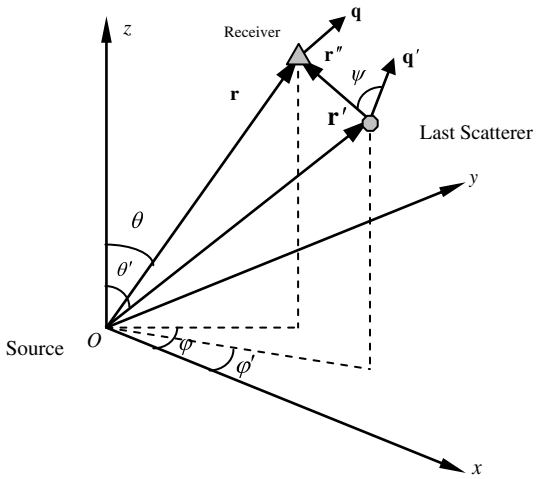


Figure 2. Geometry of the multiple nonisotropic scattering process in spherical coordinates; $\mathbf{q}$ is a vector to denote the direction of energy flux at the receiver, and $\mathbf{q}'$ denotes the direction of energy flux at the last scatter.

and

$$\begin{aligned}\bar{E}_{\text{Sato}}(\boldsymbol{\rho},\tau) &= \oint \bar{f}(\boldsymbol{\rho},\tau;\mathbf{q})d\Omega(\mathbf{q}) \\ &= \frac{1}{\rho(2\pi)^2}\int_{-\infty}^{\infty} d\omega e^{-i\omega\tau}\int_{-\infty}^{\infty} dk e^{-ik\rho}\frac{ik}{2\pi}\bar{\bar{f}}_0(k,s) \\ &= \frac{1}{\rho(2\pi)^2}\int_{-\infty}^{\infty} d\omega e^{-i\omega\tau}\frac{1}{\pi}\int_0^{\infty} dk\bar{\bar{f}}_0(k,s)k\sin k\rho;\end{aligned} \quad (8)$$

$\bar{f}(\boldsymbol{\rho},\tau;\mathbf{q})$ is the normalized directional energy density at $\boldsymbol{\rho}$ in direction $\mathbf{q}$ at time τ. $\bar{\bar{f}}_0(k,s)$ is formulated in equation (A2).

Numerical Method

According to the composition of seismic-wave energy, it was decomposed into direct wave, single scattering, and higher order (≥ 2) scattering terms:

$$E(\mathbf{r},t) = WG(\mathbf{r},t) + E^1(\mathbf{r},t) + E^M(\mathbf{r},t). \quad (9)$$

There is an analytic solution for the direct wave and single scattering term (Sato, 1995), and the discrete wavenumber method of Zeng (1993) is used to estimate the high-order scattering term $E^M(\mathbf{r},t)$. Compared with Sato's weak forward scattering example, stronger nonisotropic scattering cases can be considered in our study by including higher-order terms of the spherical harmonic series with respect to solid angles in the expansion of the directional scattering coefficient. In this case, we need to calculate higher-order Green's functions in the wavenumber–Laplace domain. Based on formulation (37) of Sato (1995), a recursive formula of Green's function is obtained:

$$\begin{aligned}\bar{\bar{G}}_l(k,s) &= -\frac{(2l-1)(s+1+\bar{b})}{lk}\bar{\bar{G}}_{l-1}(k,s) \\ &\quad + \frac{l-1}{l}\bar{\bar{G}}_{l-2}(k,s) \quad \text{for } l \geq 2,\end{aligned} \quad (10)$$

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in which k is the wavenumber, $\bar{\bar{G}}_l(k,s)$ is the Green's function for order l, its formulation is given in equation (A3), and $\bar{b} = b/V_0 g_0$ is the normalized intrinsic attenuation factor.

Data Analysis

In the numerical simulation procedure, for station j at discrete time point i, the discrete energy density is defined by $E(r_j, t_{ji})$, which can be estimated numerically by equation (9) without considering intrinsic attenuation factor b. We then introduce the intrinsic attenuation factor in exponential form and characterize the site effect by factor A_j. Thus, for event k, the energy density at station j can be expressed as

$$E_k(r_j, t_{ji}) = A_j E(r_j, t_{ji}) \exp(-b t_{ji}). \tag{11}$$

In order to fit the envelopes of the observed velocity seismogram records, we take a square root of equation (11). In calculating the velocity envelope, we assume an isotropic source radiation. For a given event k, the misfit between data and synthesis result is defined by

$$\Phi = \sum_{j=1}^{nst} \frac{1}{A_j} \sum_{i=ns}^{n} [U_{ji} - a_j f(t_{ji}) \exp(-b_v t_{ji})]^2, \tag{12}$$

in which nst is the number of stations, n is the total number of discrete time points, and ns is the number index of the moment corresponding to the S-wave arrival time. U_{ji} is the velocity envelope recorded at station j at discrete time point i, and

$$f(t_{ji}) = \sqrt{E(r_j, t_{ji})}, \qquad a_j = \sqrt{A_j}, \qquad b_v = b/2. \tag{13}$$

Given the usual range of $g_0 = 0.00005 \sim 0.3$ km^{-1}, $t_M = 0.002 \sim 0.5$ s, and we explore the best-fit parameters of g_0, t_M by using the grid-search method. The peak ratio method is used to obtain a_j. Using this method, for station j, we find the maximums of U_{ji} and $\sqrt{E(r_j, t_{ji})}$ after the S arrival time, then take the ratio between maximums of U_{ji} and $\sqrt{E(r_j, t_{ji})}$ as a_j. b_v is calculated by the least-square method, and the residual of this least-square method is expressed as

$$\lambda = \sum_{j=1}^{nst} \sum_{i=ns}^{n} (c + F_{ji} - b_v t_{ji} - Y_{ji})^2, \tag{14}$$

in which $F_{ji} = \ln[a_j f(t_{ji})]$, $Y_{ji} = \ln(U_{ji})$, and c is a modification factor that satisfies $e^c \approx 1.0$. Set $\frac{\partial \lambda}{\partial c} = 0$ and $\frac{\partial \lambda}{\partial b_v} = 0$, then c and b_v can be obtained by

$$\begin{pmatrix} c \\ b_v \end{pmatrix} = \frac{1}{nt \sum_{j=1}^{nst} \sum_{i=ns}^{n} t_{ji}^2 - \left(\sum_{j=1}^{nst} \sum_{i=ns}^{n} t_{ji}\right)^2} \begin{pmatrix} \sum_{j=1}^{nst} \sum_{i=ns}^{n} (t_{ji}) \sum_{j=1}^{nst} \sum_{i=ns}^{n} (F_{ji} t_{ji} - Y_{ji} t_{ji}) - \sum_{j=1}^{nst} \sum_{i=ns}^{n} t_{ji}^2 \sum_{j=1}^{nst} \sum_{i=ns}^{n} (F_{ji} - Y_{ji}) \\ nt \sum_{j=1}^{nst} \sum_{i=ns}^{n} (F_{ji} t_{ji} - Y_{ji} t_{ji}) - \sum_{j=1}^{nst} \sum_{i=ns}^{n} (t_{ji}) \sum_{j=1}^{nst} \sum_{i=ns}^{n} (F_{ji} - Y_{ji}) \end{pmatrix}, \tag{15}$$

in which nt is the number of discrete time points after the S-wave arrival time.

2008 Wells Earthquake

The 2008 magnitude 6.0 Wells earthquake that struck northeastern Nevada (United States) was the largest earthquake to occur in the state in the last 30 years. The area around Wells is dominated by west-dipping normal faults and corresponding east-tilted ranges (Henry and Colgan, 2011). Three major structural elements (the Ruby Mountains–East Humboldt Range, the Wells Fault, and the Mary's River and Starr Valleys) come together in the vicinity of Wells, and the interaction between them provides the regional setting of the Wells earthquake (Thorman and Brooks, 2011). The mainshock and aftershock distribution of the Wells earthquake is shown in Figure 3.

During the time of the mainshock and aftershock sequence, the USArray network was operating in Nevada. It provided a unique data set for this earthquake sequence. We collected all the seismic records of this earthquake sequence with magnitudes of 4.0–5.0 within an epicentral distance of 300 km from the Incorporated Research Institutions for Seismology Data Management Center. The aftershocks used in this study are listed in Table 1, and the distribution of stations is shown in Figure 4.

Using the Seismic Analysis Code (SAC) software (Goldstein *et al.*, 2003; Goldstein and Snoke, 2005), the instrument response errors were removed from the data. The data were filtered by Butterworth filters, and four octave passbands were selected (1–2, 2–4, 4–8, and 8–16 Hz). Then, the observed vertical envelopes are obtained by SAC. In order to fit these observed data, the simulation techniques described in the Numerical Method section are used to calculate the synthesized velocity envelopes for each frequency band based on different scattering patterns.

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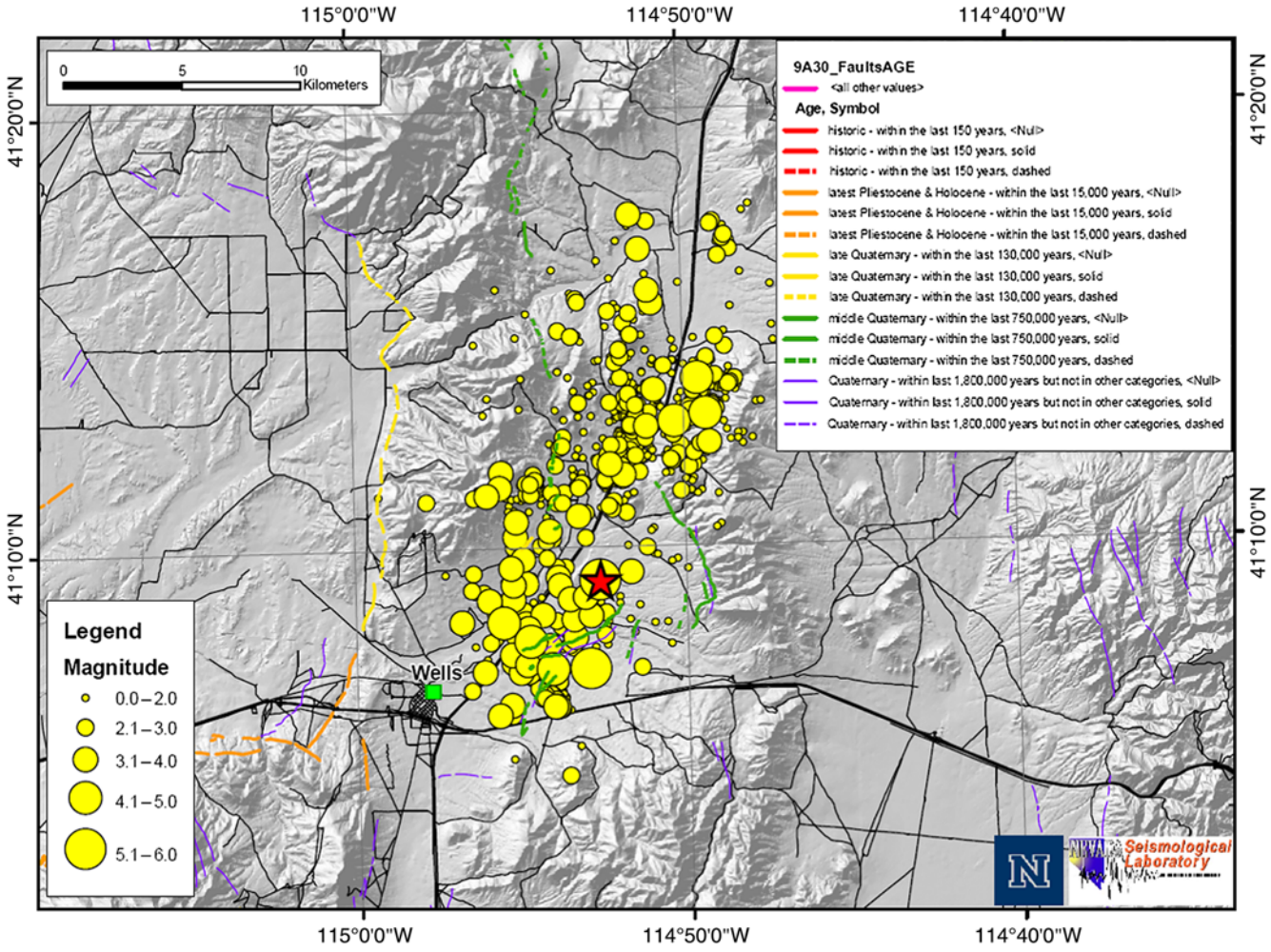


Figure 3. Location of the Febraury 2008 Wells, Nevada, earthquake sequence. The color version of this figure is available only in the electronic edition.

Introduction of Nonisotropic Scattering Pattern

In the inhomogeneous media, the wave velocity depends on the location $\mathbf{x}$ and can be described by

$$V(\mathbf{x}) = V_0\{1 + \xi(\mathbf{x})\}, \tag{16}$$

in which $\xi(\mathbf{x})$ is the fractional fluctuation of the wave velocity, $\xi(\mathbf{x}) \equiv \delta V / V_0$, and V_0 is the mean velocity $V_0 = \langle V(\mathbf{x}) \rangle$. There is an empirical linear relationship between $\xi(\mathbf{x})$ and the differential mass density of media,

$$\frac{\delta\rho(\mathbf{x})}{\rho_0} = \mu\xi(\mathbf{x}). \tag{17}$$

With increasing μ, the contributions from velocity and density fluctuations to backward scattering are rising. Using Birch's law (Birch, 1961), $V_0[\mathrm{km/s}] = 1.56\rho[\mathrm{g\,cm^{-3}}] - 0.66$, we obtain

Table 1
All Earthquakes and Their Parameters Used in This Study

Event Number	Date (yyyy/mm/dd)	Time (hh:mm:ss.sss)	Latitude (°)	Longitude (°)	Depth (km)	M_L	Station Number
1	2008/02/21	16:20:02.238	41.2045	−114.8635	5.3032	4.37	48
2	2008/02/21	23:57:51.013	41.1506	−114.9316	10.5579	4.84	40
3	2008/02/22	15:00:05.233	41.1416	−114.9230	8.4623	4.32	47
4	2008/02/22	23:27:45.268	41.1042	−114.9173	15.6331	4.71	50
5	2008/02/27	07:59:38.211	41.1851	−114.8309	8.7806	4.67	50
6	2008/04/01	13:16:17.213	41.2285	−114.8400	10.9913	4.78	43
7	2008/04/22	20:40:09.273	41.2206	−114.8063	7.5712	4.39	40

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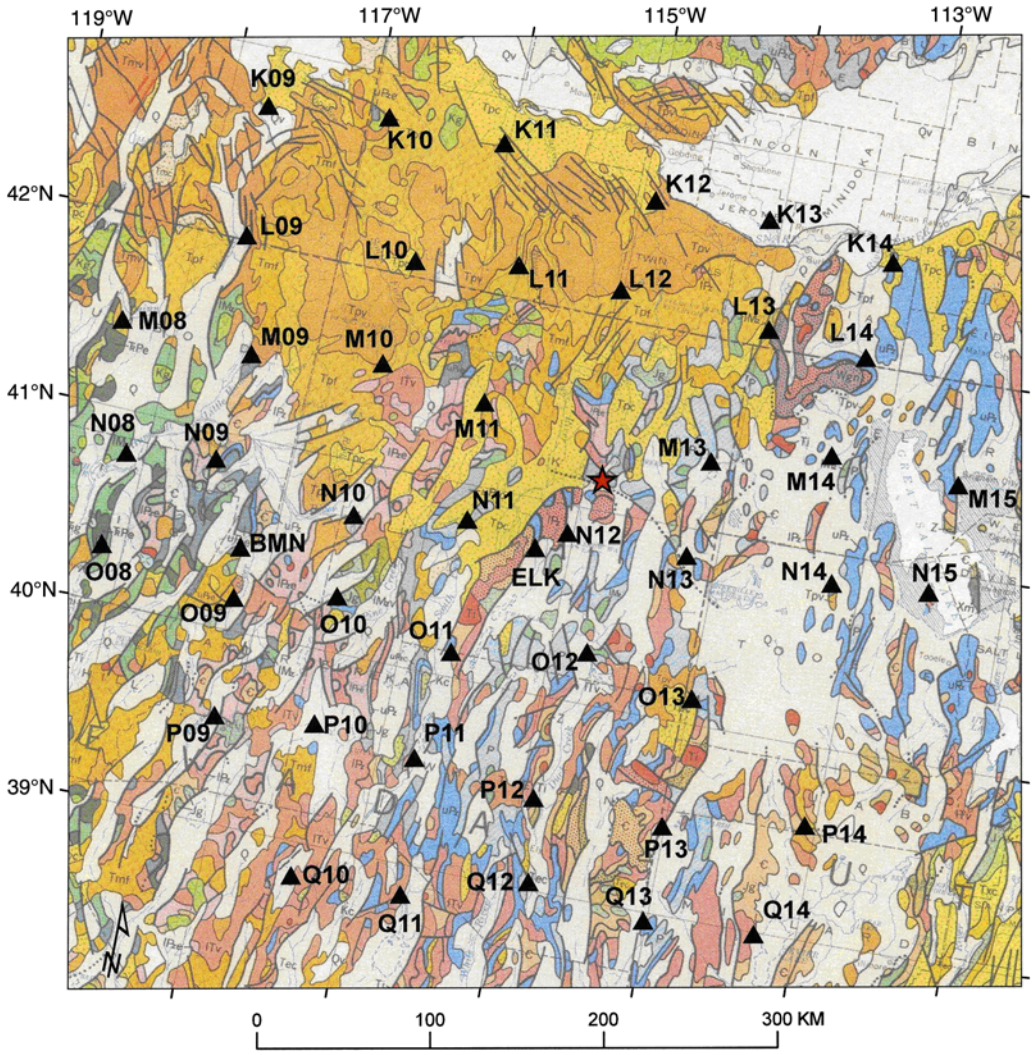


Figure 4. Distribution of stations around the 2008 Wells earthquake. Triangles are the locations of stations; the star represents the epicenter of mainshock. The color version of this figure is available only in the electronic edition.

$$\frac{\delta\rho}{\rho_0} = \left(\frac{V_0}{V_0 + 0.66}\right)\left(\frac{\delta V}{V_0}\right). \tag{18}$$

In this paper, the S-wave velocity is assumed to be $V_0 = 3.5$ km/s, corresponding to $\mu = 0.84$. In the spherical coordinates system (r, ψ, γ), as shown in Figure 5, basic backward scattering patterns are obtained by Sato and Fehler (1998)

$$X_\psi^{ss}(\psi, \gamma) = \cos\gamma[\mu(\cos\psi - \cos 2\psi) - 2\cos 2\psi] \tag{19}$$

and

$$X_\gamma^{ss}(\psi, \gamma) = \sin\gamma[\mu(\cos\psi - 1) + 2\cos 2\psi]. \tag{20}$$

Based on the Born approximation method, scattering the coefficient for S-to-S scattering is given by

$$\zeta(\psi) = \frac{k^4}{4\pi}(|X_\psi^{ss}(\psi,\gamma)|^2 + |X_\gamma^{ss}(\psi,\gamma)|^2)P\left(2k\sin\frac{\psi}{2}\right), \tag{21}$$

in which k is the wavenumber and $P(\ldots)$ is the power spectral density function (PSDF).

Using equation (21), set $\nu = 4a^2k^2$ and $\gamma = 0$, a common scattering pattern function $\zeta(\psi)$ related to exponential PSDF (Sato, 1982; Hoshiba, 1995) is introduced to describe the axial symmetric nonisotropic scattering case

$$\zeta(\psi) = \frac{1 + \nu}{[1 + \nu\sin^2(\psi/2)]^2}, \tag{22}$$

in which $\nu > 0$; that is, the forward scattering is dominated over backward scattering process. Although $\nu < 0$ is acceptable as a mathematical model, we should note that the functional form of equation (22) has an origin in the scattering pattern due to velocity fluctuation for the exponential PSDF. Thus, positive values of ν are used in this paper, and $\nu = 1.0 \times 10^{-5}$ is used to approximately denote the isotropic scattering case.

For the backward scattering case, set $\psi = \pi$ in equation (21); then, for the exponential PSDF case,

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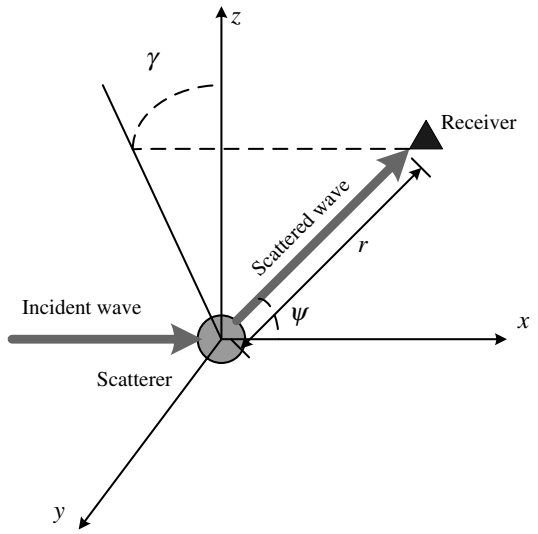


Figure 5. A schematic view of the scattering process.

$$\zeta(\psi = \pi) = \frac{8(1+\mu)^2\varepsilon^2 a^3 k^4}{(1+4a^2k^2)^2}. \tag{23}$$

When $ak \gg 1$,

$$\zeta(\psi = \pi) \approx 1.69\varepsilon^2/a, \tag{24}$$

in which ε^2/a is assumed to be 10^{-3} km^{-1} in this paper; in order to plot the misfits with the results from different scattering patterns of ν in Figure 6, a sign of $\nu = -1$ is used to denote the backward scattering pattern of equation (24).

Effective Isotropic Scattering Coefficient

When the multiple scattering process is dominant, a diffusion equation can be used to describe this scattering process, as defined in Morse and Feshbach (1953), the momentum transfer scattering coefficient is used as the effective isotropic scattering coefficient:

$$g_m = \frac{1}{4\pi}\oint(1-\cos\psi)\zeta(\psi)d\Omega(\psi)$$
$$= \frac{1}{2}\int_0^{\pi}\zeta(\psi)(1-\cos\psi)\sin\psi d\psi. \tag{25}$$

The quantity g_m is used to characterize the coda exaction in 3D random media by Gusev and Abubakirov (1996). In the diffusion regime, Sato *et al.* (2004) modeled the multiple nonisotropic scattering process as the isotropic scattering process by using an effective isotropic scattering coefficient in 2D random media having rich, short-wavelength spectra.

In this paper, g_m is obtained using equations (22) and (25), and a comparison between g_m and the Born approximation scattering coefficient is illustrated in Figure 7. A dotted line shows a circle of radius g_m, and a bold solid line denotes the scattering coefficient $\bar{\zeta}(\psi)$ predicted by the Born approximation. Here $\bar{\zeta}(\psi)$ is obtained by normalization $\bar{\zeta}(\psi) = \zeta(\psi)/4\pi$. A dashed line is used to describe

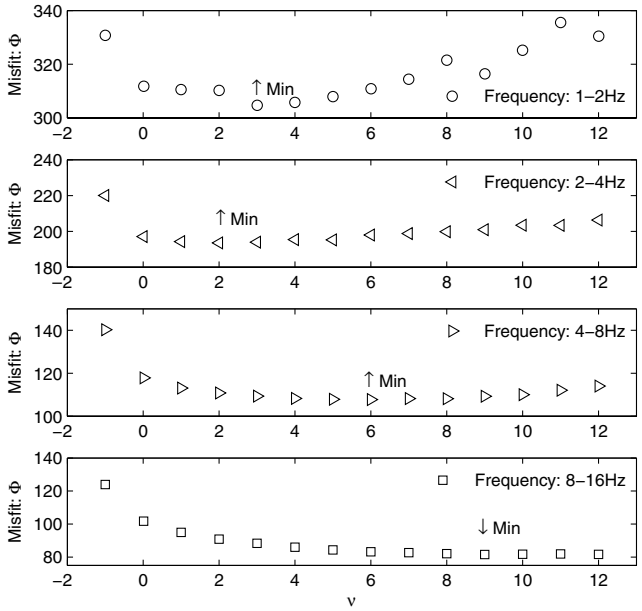


Figure 6. The change of misfit of event 4 with different scattering patterns. Arrows denote the minimum of misfits.

$(1-\cos\psi)\bar{\zeta}(\psi)$; the factor $(1-\cos\psi)$ is used to remove the forward scattering part. The dashed line is somewhat similar to the dotted line; however, there are some deviations between them that cannot be neglected. Thus, the momentum transfer scattering coefficient is not suitable to describe the multiple nonisotropic scattering process in the random media characterized by an exponential PSDF. In The Effect of Scattering Pattern on Envelope and Envelope Broadening and Peak Delay sections, as well as The Frequency Dependence of Attenuation Parameters section, the Born approximation scattering coefficient $\bar{\zeta}(\psi)$ will be used to describe the nonisotropic scattering process.

The Effect of Scattering Pattern on Envelope

To use Sato's nonisotropic scattering theory, first, $\bar{\zeta}(\psi)$ is expanded using spherical harmonic functions, as shown in equation (A4). Then, the integral kernel $\bar{f}_0(k,s)$ of equation (8) is obtained using equation (A2). In order to consider the stronger scattering case, higher-order terms of the spherical harmonic series with respect to solid angles will be included. For example, for $\nu = 9.0$, to describe this strong forward scattering pattern, at least eight spherical harmonic functions must be included in equation (A4). As shown in Figure 8, the x axis is $\bar{\zeta}(\psi)\cos\psi$, and the y axis is $\bar{\zeta}(\psi)\sin\psi$. As ν increases, the amount of forward scattering gradually increases.

Based on the series of scattering patterns for the vertical velocity seismogram of frequency bands 1–2, 2–4, 4–8, and 8–16 Hz, the model parameters of g_0 and b are obtained from the inversion of data from event 4. The degree of misfit is represented by Φ, which is obtained from equation (12). For different frequency bands, the change of misfit with

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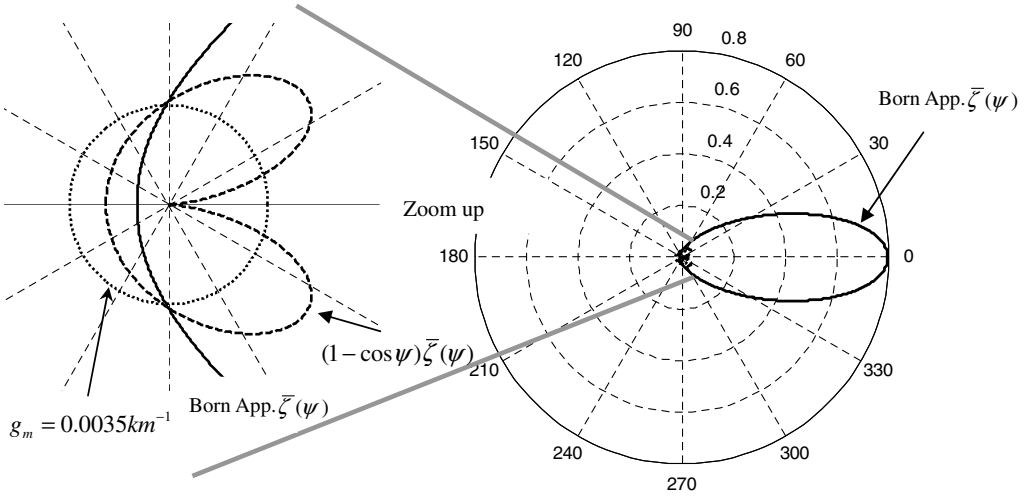


Figure 7. Comparison between the Born approximation scattering coefficient and the momentum transfer scattering coefficient.

different scattering patterns is shown in Figure 6. (Arrows denote the minimums of misfit.)

As shown in Figure 6, in general, for all frequency bands, at first, the values of misfit are decayed with increasing of ν, then the minimums of misfit will reach in the region of $\nu > 0$, and after the minimums, there will be a slight increase of the values of misfit with the increase of ν. Hence, as the scattering process gradually changes from backward to isotopic and then to forward-dominated scattering, the envelope fitting will become more accurate, and then the fitting results will become slightly poor again in the region of $\nu > 0$. However, it is obvious that all the minimums of Φ locate at the region of $\nu > 0$, which means that best-fitting results are calculated from forward-dominated scattering pattern. Thus, we conclude that, compared with backward scattering and isotropic scattering patterns, the forward-dominated scattering pattern can better fit the high-frequency observed data.

To check the variation of synthesized envelopes, which are calculated from backward, forward, and isotropic scattering processes, equation (24), $\nu = 1.0 \times 10^{-5}$, and $\nu = 9.0$ are picked to represent the backward, isotropic, and forward scattering patterns, respectively. Based on these, synthesized envelopes are obtained (Fig. 9), and four stations are selected to show the difference between synthesized envelopes from different scattering patterns. The hypocentral distance of these four stations is changed from 77 km to 283 km. From the comparisons between synthesized envelopes and observed data, we found that forward scattering pattern can fit the data much better than the backward and isotropic scattering case for all different hypocentral receivers. Here, V_b, V_i, and V_f are used to denote peak values of the envelopes, which are calculated from backward, isotropic, and forward scattering patterns, respectively. For all the stations, the value of V_b is higher than V_i and V_f. At smaller hypocentral distance receivers (e.g., station N11), $V_f > V_i$. However, with the increase of hypocentral distance, the deviation between them will be decreased. Take station M10 and P10 for example, the deviations are almost disappeared, that is $V_f \approx V_i$. When the hypocentral distance increases, this trend will reverse; for example, it turns into $V_i > V_f$ for station Q10. For the early coda wave (after the peak value time and before the middle coda wave), all the results from the three patterns agree well with each other and exhibit a good fitness with the observed data. However, around the middle coda part, except in the forward scattering case, both isotropic and backward scattering results failed to predict the data. We use V_{bm}, V_{im}, and V_{fm} to represent the middle coda value of the envelope, which is calculated from backward, isotropic, and forward scattering patterns, respectively. It evident that, for smaller hypocentral distance receivers such as station N11, $V_{bm} > V_{im} > V_{fm}$. The difference between V_{bm}, V_{im}, and V_{fm} decreased as the hypocentral distance increased for station M10 ($V_{bm} \approx V_{im} > V_{fm}$), and it is $V_{bm} < V_{im} \approx V_{fm}$ for station P10. For the larger hypocentral distance, the deviation between different scattering patterns becomes more obvious; for instance, for station Q10, $V_{bm} < V_{im} < V_{fm}$. For the later coda, the envelopes from all three scattering patterns agree well with observed data, and all of them gradually decay to zero. As stated previously in this paper for the comparisions, a significant difference is apparent in the envelopes, as calculated by backward, isotropic, and forward scattering patterns. As the hypocentral distance increased, the forward scattering pattern can transfer the direct wave energy to the middle coda wave part more efficiently than others. Furthermore, the envelopes from the forward scattering pattern can capture the observed data from the *S*-arrival time to a later coda. Hence, it is more reasonable that high-frequency *S*-coda waves originate from the leakage of energy from the multiple forward-dominated scattering process, instead of the backward or isotropic scattering processes.

In Appendix B, for data from event 4, more comparisons between envelopes are obtained from the forward scattering pattern of $\nu = 9.0$, and the observed data are listed. From the early direct *S*-wave group to the later coda, the synthesized

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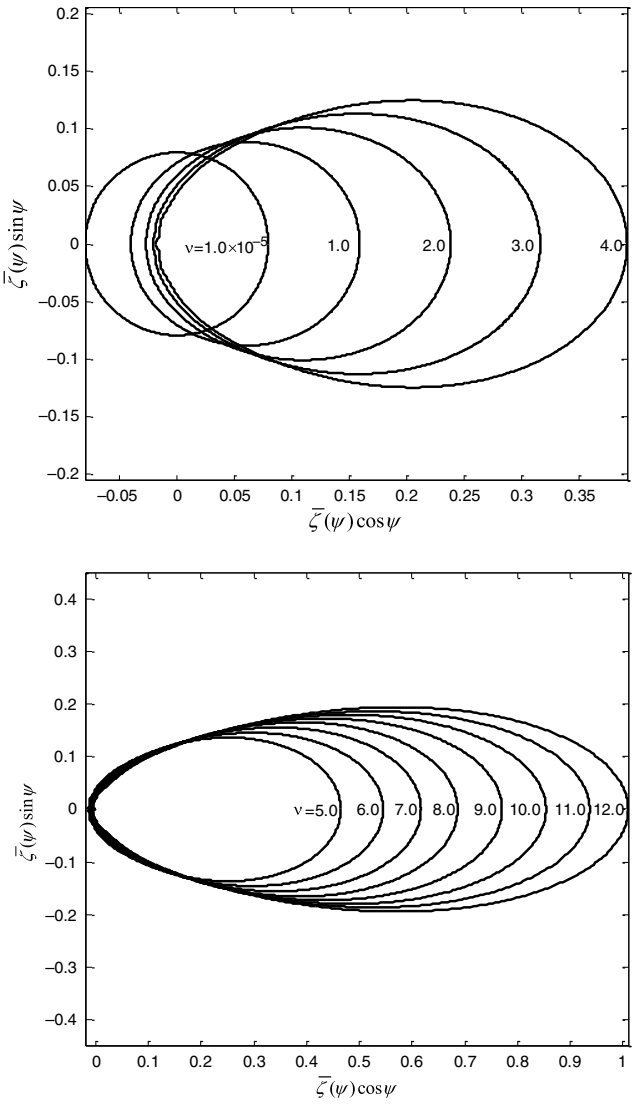


Figure 8. Nonisotropic scattering patterns.

envelope exhibits a reasonable goodness of fit for almost all 50 stations. The fitting results of 18 representative stations are plotted in Figure B1. Other stations with redundant hypocentral distances are not selected for plotting.

Moreover, to investigate the effect of t_M on the synthesized envelope, using the forward scattering patterns shown in Figure 6 (marked by "Min"), the initial MS envelopes are synthesized for different values of t_M at different frequencies. As shown in Figure 10, the MS envelope duration increased with the increased t_M. Meanwhile, the envelope decayed more slowly. Hence, t_M is a parameter to describe the duration of the initial MS envelope at the source.

From the inversion of the seven events listed in Table 1, the optimal values of t_M are obtained for different frequency bands. As shown in Figure 11, the bold solid curve denotes the mean value of t_M, which decreased when frequency increased. The mean values of t_M are 0.12, 0.14, 0.22, and 0.29 for the 12, 6, 3, and 1.5 Hz *S* waves, respectively.

Envelope Broadening and Peak Delay

A seismic wave is initially an impulse at the source, and its duration becomes longer as travel distance increases. This increasing wave packet duration is called envelope broadening. The broadening of high-frequency *S*-wave envelopes has been reported for local earthquakes (Sato, 1989; Scherbaum and Sato, 1991; Petukhin and Gusev, 2003), and envelope broadening was only investigated by a stochastic treatment of parabolic approximation of the wave equation (Sato, 1989; Saito *et al.*, 2002; Takahashi *et al.*, 2009). In this section, using the vertical velocity seismograms from event 4, with frequency band 8–16 Hz, the broadening of the velocity envelope is discussed based on envelopes calculated from the nonisotropic scattering process. Equation (24), $\nu = 1.0 \times 10^{-5}$, and $\nu = 9.0$ are used to represent backward, isotropic, and forward scattering patterns, respectively. Based on these patterns, the velocity envelopes are synthesized for different stations.

In Figure 12, t_P is the time duration from the *S*-wave arrival time to envelope peak time, and t_Q is the time duration from the *S*-wave arrival time to the later half-peak of the envelope. According to the measurements of Sato (1989), the envelope broadening can be described by plotting t_P, t_Q versus hypocentral distance r. For different stations, t_P, t_Q of the data and synthesized envelope, respectively, are picked, then the discrete values of t_P, t_Q from the synthesis result and observation data are fitted by linear regression as follows:

$$\log_{10} t_P[\mathrm{s}] = (A_P) \log_{10} r[\mathrm{km}] + (B_P),$$
$$\log_{10} t_Q[\mathrm{s}] = (A_Q) \log_{10} r[\mathrm{km}] + (B_Q) \tag{26}$$

and

$$\log_{10} t_P[\mathrm{s}] = (A_{OP}) \log_{10} r[\mathrm{km}] + (B_{OP}),$$
$$\log_{10} t_{OQ}[\mathrm{s}] = (A_Q) \log_{10} r[\mathrm{km}] + (B_{OQ}), \tag{27}$$

in which (A_P, B_P, A_Q, B_Q) and (A_{OP}, B_{OP}, A_{OQ}, B_{OQ}) denote the regression coefficients of synthesized result and observed data, respectively.

As shown in Figure 13, linear regression results from both the backward and isotropic scattering appear to be flat. As a result, there are almost no broadening phenomena in their synthesized envelopes. In contrast, the results obtained by the forward scattering pattern are nearly the same as the observed data. Both t_P and t_Q are increased with the rising of hypocentral distance. Also, compared with the gradient of the linear regression of data, the forward scattering pattern fit much better than did the isotropic and backward scattering cases. Therefore, the multiple forward scattering model is able to predict the peak delay and broadening of the seismogram envelopes. We concluded that high-frequency *S*-wave seismogram envelope broadening can be explained by the seismic-wave multiple forward-dominated scattering process that occurred in the inhomogeneous media. This conclusion is in agreement with Sato (1989).

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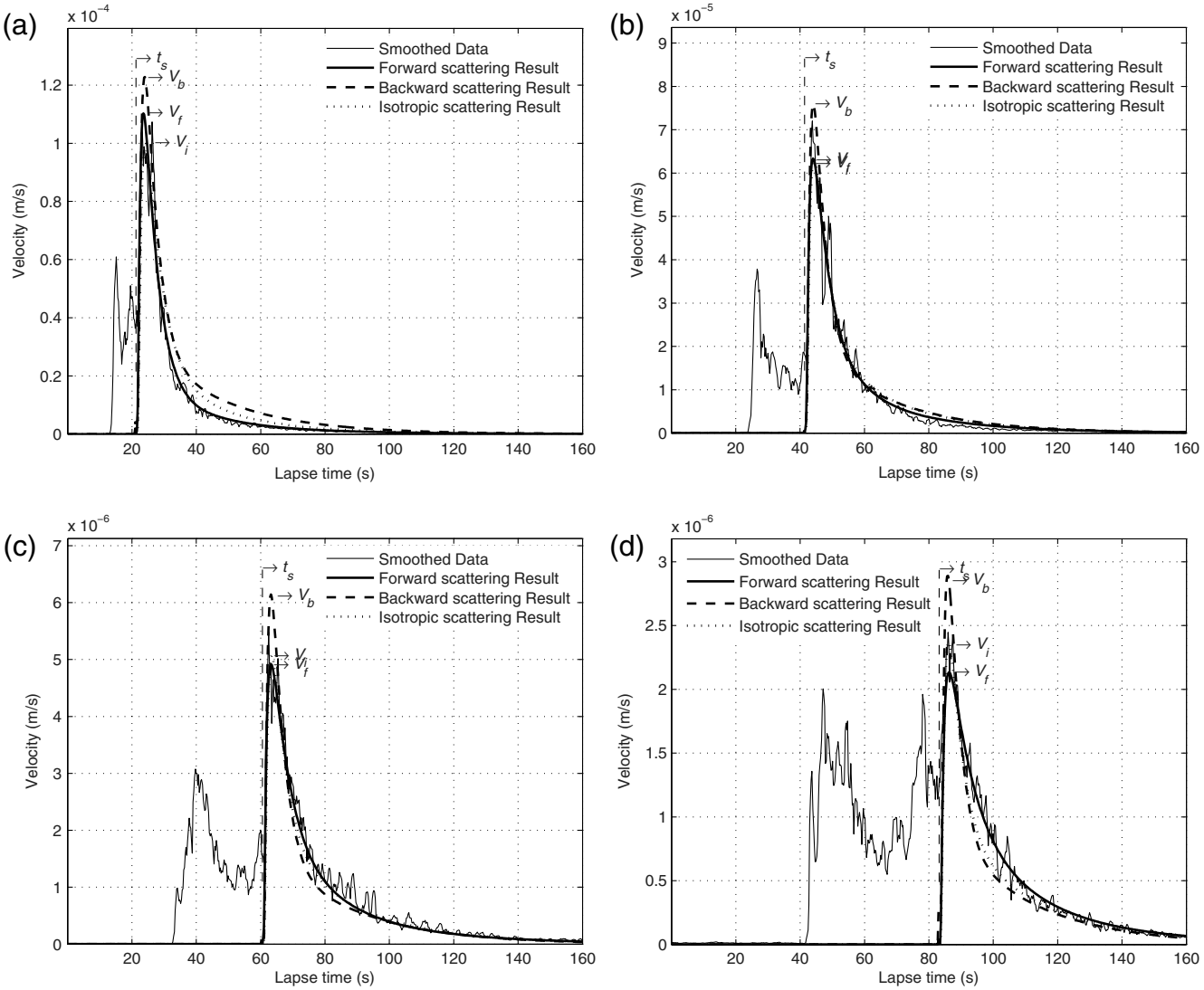


Figure 9. Numerical comparisons between observed data and the backward (equation 24), isotropic ($\nu = 1.0 \times 10^{-5}$), and forward ($\nu = 9.0$) scattering results. The travel distances of the seismic wave are (a) 77 km (station N11), (b) 144 km (station M10), (c) 211 km (station P10), and (d) 283 km (station Q10), respectively.

Given a set of t_M and b, based on the forward scattering pattern of $\nu = 9.0$, the MS envelopes are synthesized by solving equation (6), the envelope duration t_Q is picked, and we plot t_Q against t_M in Figure 14, assuming

$$\log_{10}(t_Q)[\mathrm{s}] = A_{qm} \log_{10}(t_M)[\mathrm{s}] + B_{qm}. \qquad (28)$$

From the numerical results in Figure 14, under different intrinsic attenuation factor b, the relationship between t_M and t_Q can be described reasonably well by equation (28). With an increase in b, slight variations appear in A_{qm} and B_{qm}. Thus, t_Q is a parameter that is insensitive to the intrinsic attenuation factor of the media, and therefore it is reliable to measure the envelope broadening by using t_Q.

Finally, in Figures 15 and 16, resultant parameters A_P, B_P, A_Q, and B_Q are compared with other regions in the world. KIB, OHR, ASO, SHM, YMK, OSM, JIZ, KGN, and TOE are stations in the Kanto–Tokai (KT) area of Japan (Obara and Sato, 1995). Data for the southeastern (SE) Honshu area, Japan, is from Sato (1989); Changbai (CB) Mountain, China, is from Fan *et al.* (2009), and the northeastern (NE) Honshu area is from Saito *et al.* (2002). Our observation data is from the Wells area, Nevada (NV), and this study provides the synthesized results obtained by forward scattering patterns (marked by "Min" in Fig. 6). It should be noted that focal depths are limited to be larger than 35 km in most analysis results of Japan, but all focal depths of the events used in this paper are shallow crustal earthquakes. The former analysis (Sato, 1989; Obara and Sato, 1995; Saito *et al.*, 2002), is mainly focused on the envelope of *S* waves especially in the early part. For shallow earthquakes, head waves appear in advance of the direct wave, which modifies the envelope form, especially for distances larger than 100 km. In continental regions, there are

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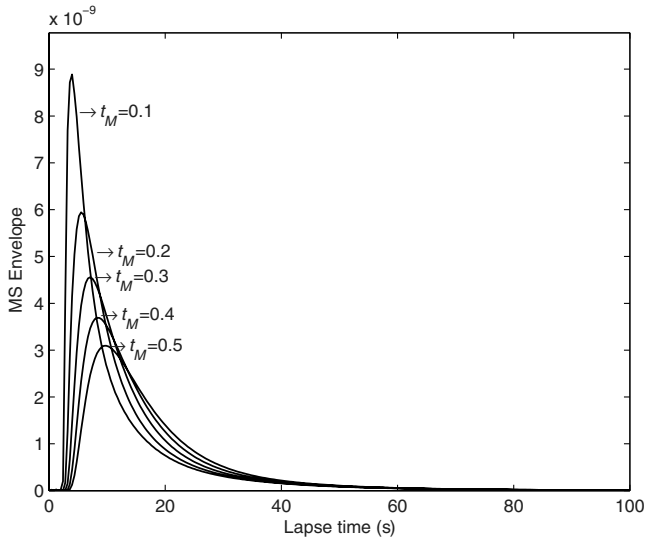


Figure 10. The effect of t_M on synthesized initial mean square (MS) envelope.

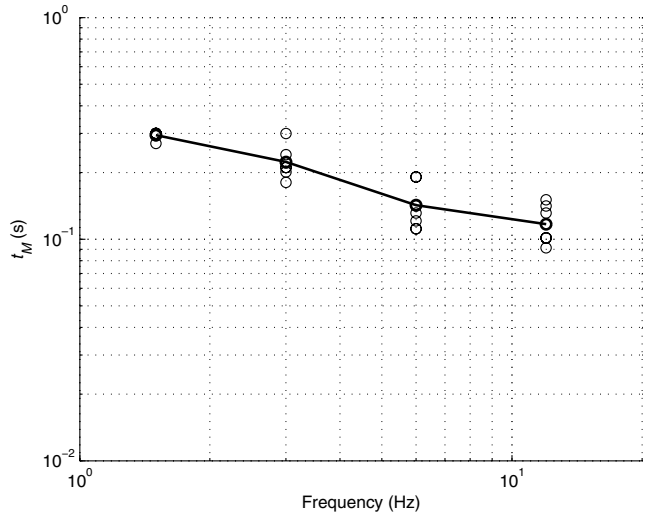


Figure 11. The change of t_M with frequency.

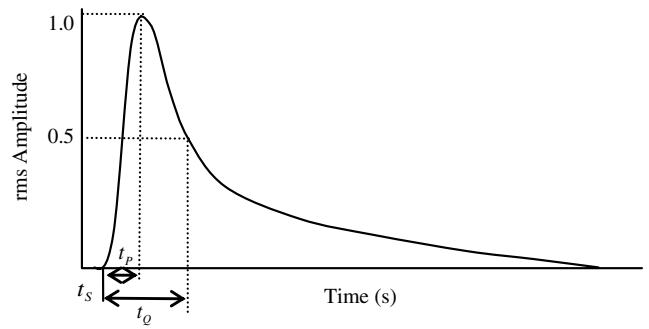


Figure 12. Illustration of t_P, t_Q of velocity envelope.

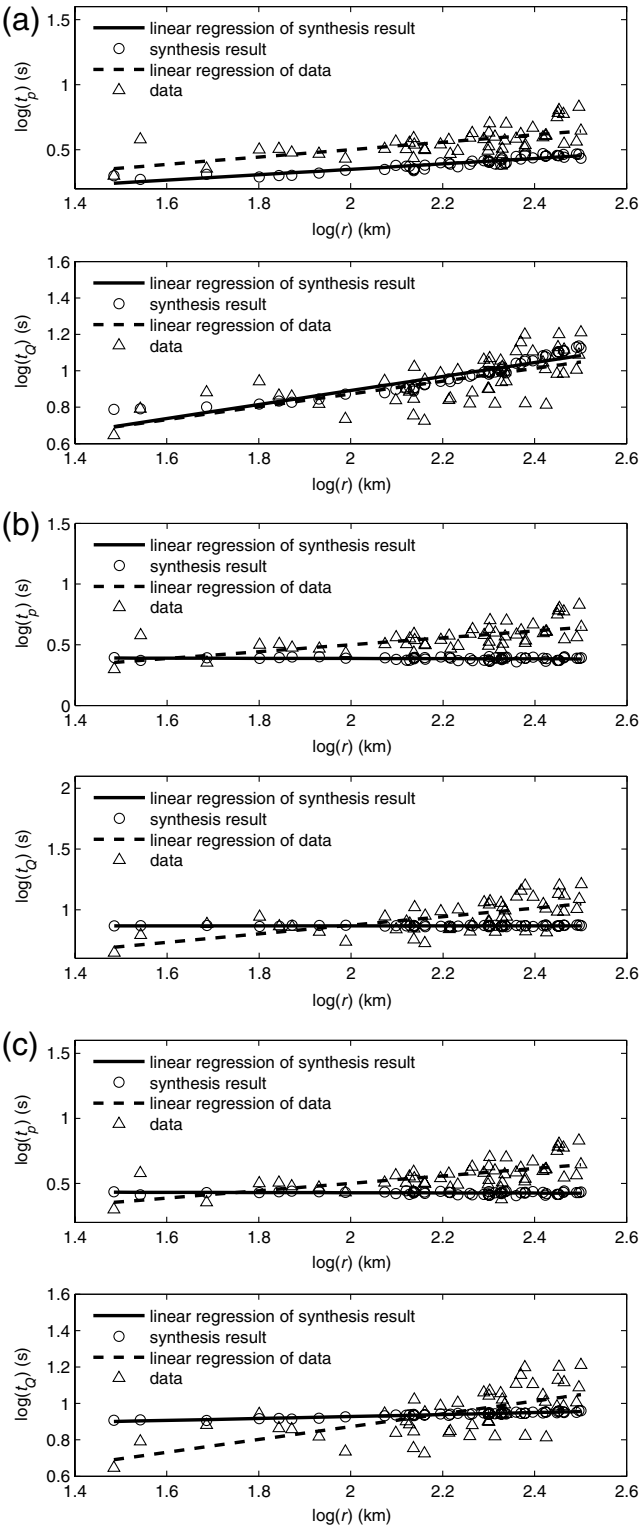


Figure 13. t_P, t_Q versus hypocentral distance, log means base 10 logarithm. (a) Forward scattering result ($\nu = 9.0$), $A_P = 0.21$, $B_P = -0.07$, $A_{OP} = 0.28$, $B_{OP} = -0.07$; $A_Q = 0.38$, $B_Q = 0.12$, $A_{OQ} = 0.35$, and $B_{OQ} = 0.17$. (b) Backward scattering result (equation 24), $A_P = -0.0068$, $B_P = 0.40$, $A_{OP} = 0.28$, $B_{OP} = -0.07$; $A_Q = 0.0008$, $B_Q = 0.87$, $A_{OQ} = 0.35$, $B_{OQ} = 0.17$. (c) Isotropic scattering result ($\nu = 1.0 \times 10^{-5}$), $A_P = -0.007$, $B_P = 0.44$, $A_{OP} = 0.28$, $B_{OP} = -0.07$; $A_Q = 0.05$, $B_Q = 0.82$, $A_{OQ} = 0.35$, and $B_{OQ} = 0.17$.

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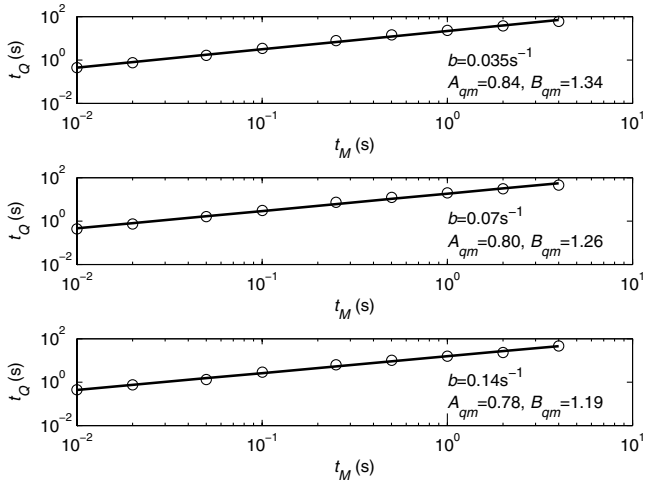


Figure 14. Plots of envelope duration t_Q (circles) against t_M for various intrinsic attenuation factors b. The solid line is the regression line given by equation (28).

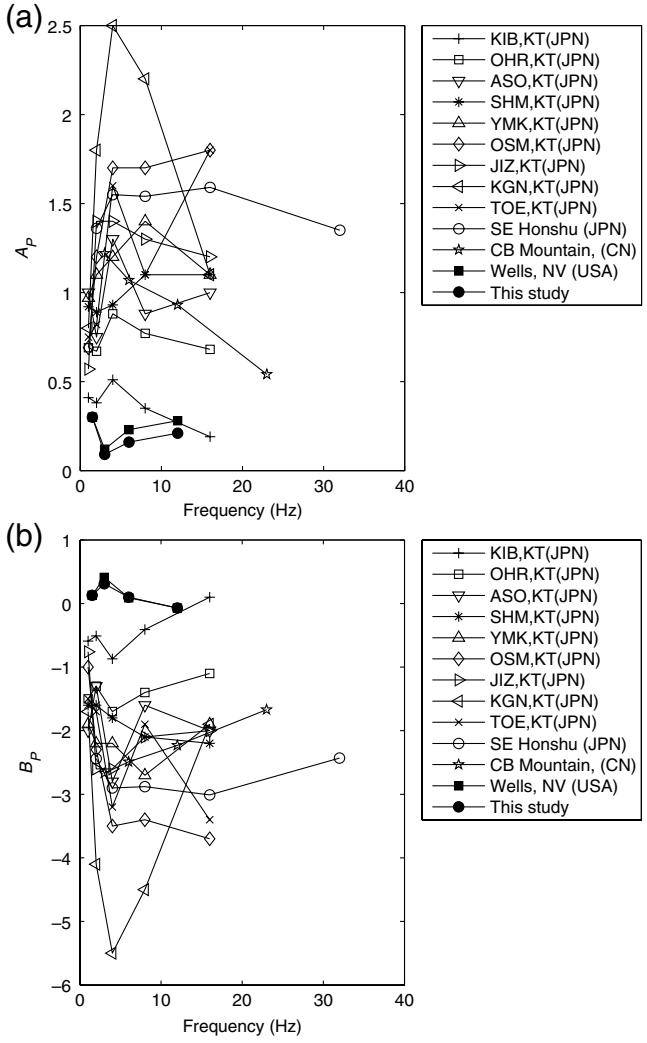


Figure 15. Comparison of A_P, B_P versus frequency for different regions in the world.

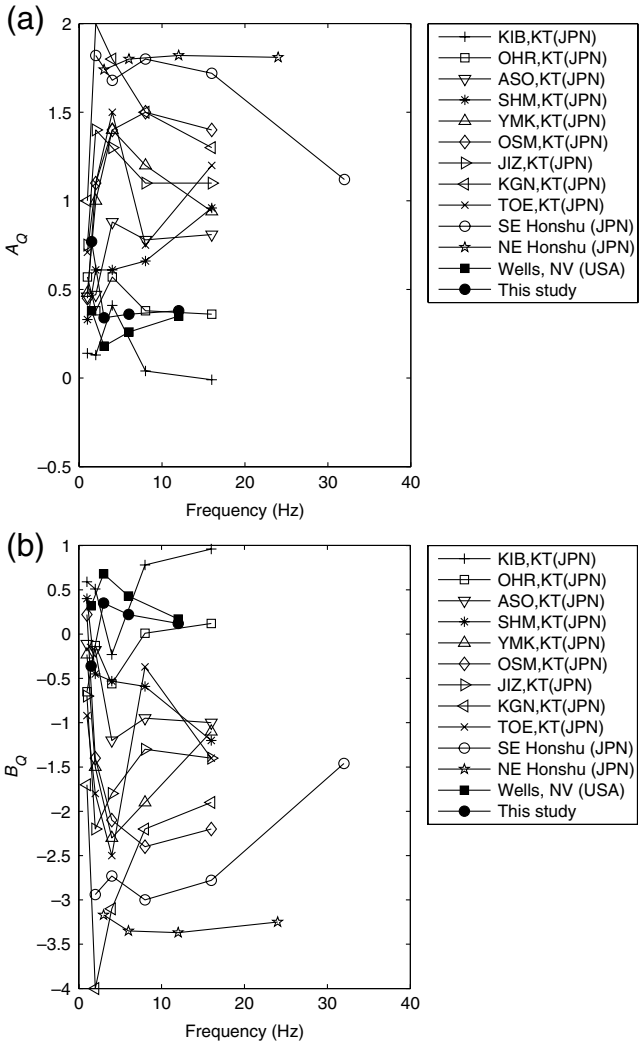


Figure 16. Comparison of A_Q, B_Q versus frequency for different regions in the world.

contributions of multiple reflection as *Lg* waves. As shown in Figures 15 and 16, our values of A_P are small and the values of B_P are large compared with those in Japan. Differences are obvious. It means that our envelopes are broadening well, even at a distance of 40 km, which might reflect the contribution of *Lg* waves. Generally speaking, however, both A_P and A_Q are larger than zero, which means the envelope broadening is apparent in all regions. Also, for higher frequencies (>5 Hz), the change in these parameters with frequency is slight in most regions.

The Frequency Dependence of Attenuation Parameters

When a seismic wave propagates through the inhomogeneous Earth medium, its amplitude generally decays with increasing travel distance due to intrinsic and scattering attenuation. The attenuation parameters of high-frequency waves are fascinating because of their sensitivity to the details of seismic source and the Earth's structure, and many

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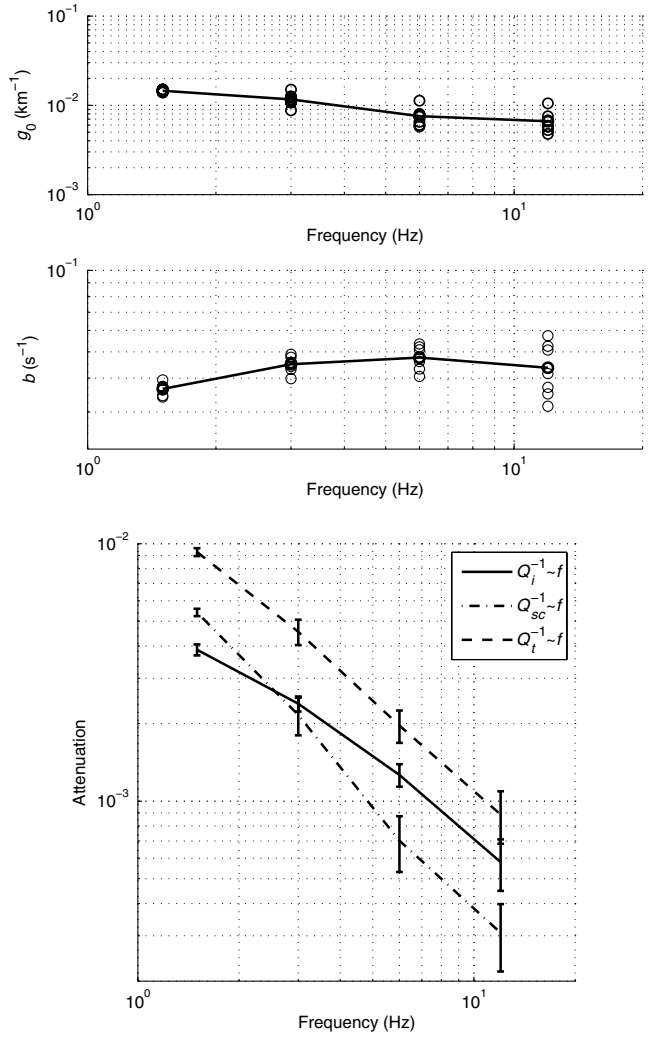


Figure 17. Frequency dependence of (a) scattering coefficient g_0, (b) intrinsic attenuation factor b, and (c) scattering attenuation Q_{Sc}^{-1}, intrinsic attenuation Q_I^{-1}, and total attenuation Q_t^{-1}. The error bar represents one standard deviation level of average value of Q_I^{-1}, Q_{Sc}^{-1}, and Q_t^{-1}. The circles represent results from seven different events; thick circles represent the average value.

methods have been developed to estimate the attenuation parameters. For example, the radiative transfer theory has been used to obtain the separated estimation of the scattering attenuation and intrinsic attenuation (Wu, 1984; Hoshiba, 1993), and, based on multiple lapse time-window method, many seismologists have studied the frequency dependence of attenuation parameters in different regions (Fehler *et al.*, 1992; Hoshiba, 1993).

In this section, based on the multiple forward-dominated scattering pattern, the scattering attenuation Q_{Sc}^{-1} and intrinsic attenuation Q_I^{-1} of high frequencies are estimated separately, and their frequency dependence is discussed. First, vertical velocity seismograms are selected from the seven events listed in Table 1, and four octave frequency bands (8–16, 4–8, 2–4, and 1–2 Hz) are used to band-pass filter the data. Accordingly, their central frequencies are 12, 6, 3, and 1.5 Hz, respectively. Then, on the basis of the forward-dominated scattering patterns listed in Figure 6 (marked by "Min"), synthesized envelopes are obtained. By fitting the observed data, optimal g_0 and b are obtained for the seven different events. Finally, scattering attenuation, intrinsic attenuation, and total attenuation are calculated by

$$\begin{aligned} Q_I^{-1} &= b/2\pi f, \\ Q_{Sc}^{-1} &= V_0 g_0/2\pi f, \\ Q_t^{-1} &= Q_I^{-1} + Q_{Sc}^{-1}. \end{aligned} \tag{29}$$

In Figure 17a, the total scattering coefficient g_0 varies between 0.005 km^{-1} and 0.015 km^{-1}, a slight decay of the average value of g_0 is apparent with the increase in frequency. In Figure 17b, the average value of intrinsic attenuation factor b increases with frequency before 6 Hz and then decreases with frequency; it ranges from 0.015 to 0.025. The frequency dependence of Q_I^{-1}, Q_{Sc}^{-1} and total attenuation Q_t^{-1} is illustrated in Figure 17c, in which Q_I^{-1}, Q_{Sc}^{-1} and Q_t^{-1} decrease when frequency increases. Intrinsic attenuation is dominant over scattering attenuation at higher frequencies (2–16 Hz), and scattering attenuation is higher than intrinsic attenuation only for the lowest-frequency band (1–2 Hz); this result is agreement with the western region of India (Ugalde *et al.*, 2007).

In Figure 18, resultant Q_I^{-1}, Q_{Sc}^{-1} values are compared with those values in other regions. From these comparisons, it is apparent that both Q_I^{-1} and Q_{Sc}^{-1} decreased when frequency increased. The overall trend of scattering attenuation is similar to those in southern California, and the frequency dependency of intrinsic attenuation is close to that in the Hokkaido area of Japan.

Conclusion and Discussion

Based on Sato's formulation of multiple nonisotropic scattering theory, a characteristic source time is used to define the initial energy density impulse duration at the source, and an analytical representation of intensity spectral density of initial spherical radiated impulse wave is incorporated into Sato's energy density integral equation. A recursive scheme is used to calculate the higher-order Green's functions, so that stronger nonisotropic scattering patterns can be used in the inversion of velocity envelopes of the aftershocks of the 2008 Wells, Nevada, earthquake.

First, a scattering pattern function is used to generate different nonisotropic scattering patterns. Based on those patterns, velocity envelopes are synthesized. Our best-fit results show $\nu > 0$ for all seven aftershocks and suggests forward scattering dominates the scattering process. In addition, there is a significant difference between the synthesized envelopes, which are estimated by backward, isotropic, and forward scattering patterns, respectively. For example, for small hypocentral distance receivers, envelopes calculated by these patterns show good agreement around the

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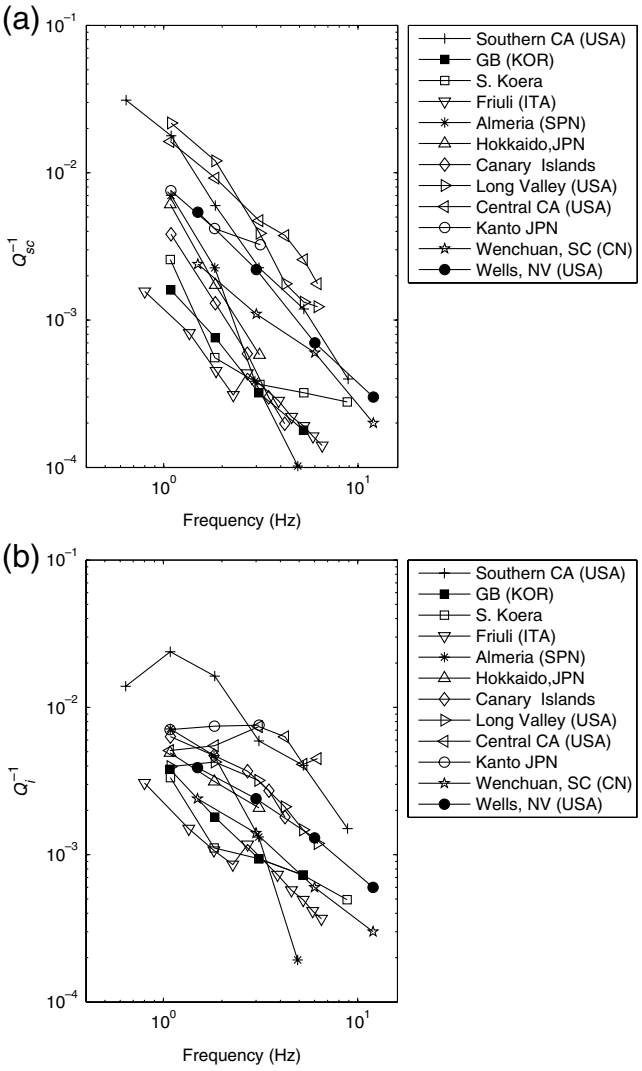


Figure 18. Comparison of Q_s^{-1}, Q_i^{-1} versus frequency for different regions in the world: southern California (Jin *et al.*, 1994); Gyeongsang Basin (GB), South Korea (Lee *et al.*, 2010); South Korea (Chung and Yun, 2009); Friuli (Bianco *et al.*, 2005); Almeria (Pujades *et al.*, 1997); Canary Islands (Canas *et al.*, 1998); Long Valley, Central California (Mayeda *et al.*, 1992); Hokkaido, Kanto (Hoshiba, 1993); Wells, Nevada (this study).

early *S*-coda wave; however, after that the envelope values from the backward-dominated scattering pattern gradually become higher than in the isotropic scattering case, and both of them present a slower decay than the forward scattering case, which provide the best fitness with observed data. Thus, from the numerical comparison with data, high-frequency *S*-coda wave can be derived from the energy leaked from the multiple forward-dominated scattering process, instead of the backward or isotropic scattering processes in the inhomogeneous media.

Furthermore, through the discussion in the Envelope Broadening and Peak Delay section comparing envelopes of the isotropic and backward scattering patterns and based on the multiple forward scattering pattern, the synthesized envelopes can reproduce the peak delay and envelope broadening of the high-frequency seismogram. Thus, in addition to the parabolic approximation of the wave equation, the forward scattering model also can be used to investigate the envelope broadening. The same conclusion is made as in Sato (1989): the high-frequency seismogram envelope broadening is due to the *S*-wave multiple forward scattering process in the media.

Finally, based on the radiative transfer theory, scattering attenuation can be separated from intrinsic attenuation. From the synthesized envelope, in the 1–16 Hz frequency range, both scattering attenuation and intrinsic attenuation decrease when frequency increases. At high frequencies (2–16 Hz), intrinsic attenuation dominates over scattering attenuation. This same conclusion has been generally reached for observations in other places over the world (Padhy *et al.*, 2007; Ugalde *et al.*, 2007; Carcole and Sato, 2010; Lee *et al.*, 2010; Mukhopadhyay *et al.*, 2010). Therefore, the scattering mechanism observed during the aftershocks of Wells, Nevada, earthquake could be more general, and the high-frequency seismic coda waves originated from scattered energy leaking out of the multiple forward scattering processes.

Data and Resources

The seismograms used in this article were collected by a network of the USArray. Data can be obtained from the Incorporated Research Institutions for Seismology Data Management Center at www.iris.edu (last accessed July 2008).

Acknowledgments

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Appendix A

Equations Cited from Sato (1995) and Errata in Sato (1995) as Corrected in Sato and Fehler (1998)

The delta function for solid angles is defined as

$$\begin{aligned}\delta_\Omega(\mathbf{r};\mathbf{r}') &= \delta_\Omega(\theta,\varphi;\theta',\varphi') \\ &= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} Y_{lm}(\theta,\varphi)Y^*_{lm}(\theta',\varphi') \\ &= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} Y^*_{lm}(\theta,\varphi)Y_{lm}(\theta',\varphi'),\end{aligned} \tag{A1}$$

in which $\mathbf{r} = (1,\theta,\varphi)$ and $\mathbf{r}' = (1,\theta',\varphi')$ are unit vectors in spherical coordinates. $Y_{lm}(\theta,\varphi)$ is the normalized spherical harmonic function, and $Y^*_{l,m}(\theta,\varphi) = (-1)^{l-m}Y_{l,-m}(\theta,\varphi)$.

The integral kernel $\bar{\bar{f}}_0(k,s)$ of equation (8) is obtained by $\bar{\bar{f}}_l(k,s)|_{l=0}$

$$\begin{aligned}\bar{\bar{f}}_l(k,s) = \bar{\bar{G}}_l(k,s) + \frac{4\pi}{2l+1}\sum_{l'=0}^{\infty}\sqrt{\frac{4\pi}{2l'+1}}\times\bar{\zeta}_{l'}\sum_{m'=-l'}^{l'}\sum_{l''=0}^{\infty}(-i)^{l'+l''-l} \\ \times\sum_{m''=-l''}^{l''}|(Y_{l'',m''})^{l,m'+m''}_{l',m'}|^2\bar{\bar{G}}_{l''}(k,s)\bar{\bar{f}}_{l'}(k,s),\end{aligned} \tag{A2}$$

in which

$$\begin{aligned}\bar{\bar{G}}_l(k,s) &= \int_0^\infty j_l(k\rho)e^{-(s+1)\rho}d\boldsymbol{\rho} \\ &= \frac{1}{k}\left(\frac{k}{2(s+1)}\right)^{l+1}\frac{\sqrt{\pi}\Gamma(l+1)}{\Gamma(l+3/2)^2} \\ &\quad\times F_1\left[\frac{l+1}{2},\frac{l+2}{2},l+\frac{3}{2},-\left(\frac{k}{s+1}\right)^2\right].\end{aligned} \tag{A3}$$

$\bar{\zeta}_l$ are coefficients of the spherical harmonic functions' expansion of the normalized scattering pattern function $\bar{\zeta}(\psi)$, which can be formulated as

$$\begin{aligned}\bar{\zeta}(\psi) &= \sum_{l=0}^{\infty}(-i)^l\bar{\zeta}_l Y_{l,0}(\cos^{-1}(\mathbf{r}''\cdot\mathbf{q}'),0) \\ &= \sum_{l=0}^{\infty}\sqrt{\frac{4\pi}{2l+1}}\bar{\zeta}_l\sum_{m=-l}^{l}Y_{lm}(\theta'',\varphi'')Y^*_{lm}(\theta_{q'},\varphi_{q'}),\end{aligned} \tag{A4}$$

in which $\bar{\zeta}(\psi) = \zeta(\psi)/4\pi$, $\mathbf{r}'' = (1,\theta'',\varphi'')$, and $\mathbf{q}' = (1,\theta_{q'},\varphi_{q'})$ are unit vectors in spherical coordinates, as shown in Figure 2.

$(Y_{lm})^{l_1m_1}_{l_2m_2}$ in equation (A2) is the product of three spherical harmonic functions, which are written using Wigner 3-j symbols:

$$\begin{aligned}(Y_{lm})^{l_1m_1}_{l_2m_2} &\equiv \oint Y^*_{l_1m_1}(\theta,\varphi)Y_{lm}(\theta,\varphi)Y_{l_2m_2}(\theta,\varphi)d\Omega \\ &= (-1)^{m_1}i^{l-l_1+l_2}\begin{pmatrix} l_1 & l & l_2 \\ -m_1 & m & m_2\end{pmatrix}\begin{pmatrix} l_1 & l & l_2 \\ 0 & 0 & 0\end{pmatrix} \\ &\quad\times\sqrt{\frac{(2l_1+1)(2l+1)(2l_2+1)}{4\pi}}.\end{aligned} \tag{A5}$$

According to the selection rules corresponding to the addition of angular momentums, this integral vanishes except when $m_1 = m + m_2$ and $|l-l_2| \le l_1 \le |l+l_2|$. Then $l + l_1 + l_2$ is even, Wigner 3-j symbols are pure real. The square of equation (A4) is

$$\begin{aligned}|(Y_{lm})^{l_1m_1}_{l_2m_2}|^2 &= \frac{1}{4\pi}\begin{pmatrix} l_1 & l & l_2 \\ -m_1 & m & m_2\end{pmatrix}^2 \\ &\quad\times\begin{pmatrix} l_1 & l & l_2 \\ 0 & 0 & 0\end{pmatrix}^2(2l_1+1)(2l+1)(2l_2+1).\end{aligned} \tag{A6}$$

Appendix B

Figure B1 compares the observed root mean square (rms) velocity amplitude envelopes (thin curve) with their corresponding synthesized envelopes (thick curve) at various stations for the 8–16 Hz frequency band for event 4 in Table 1. t_s represents S arrival time. The comparison figures are listed in an ascending order of hypocentral distances of stations.

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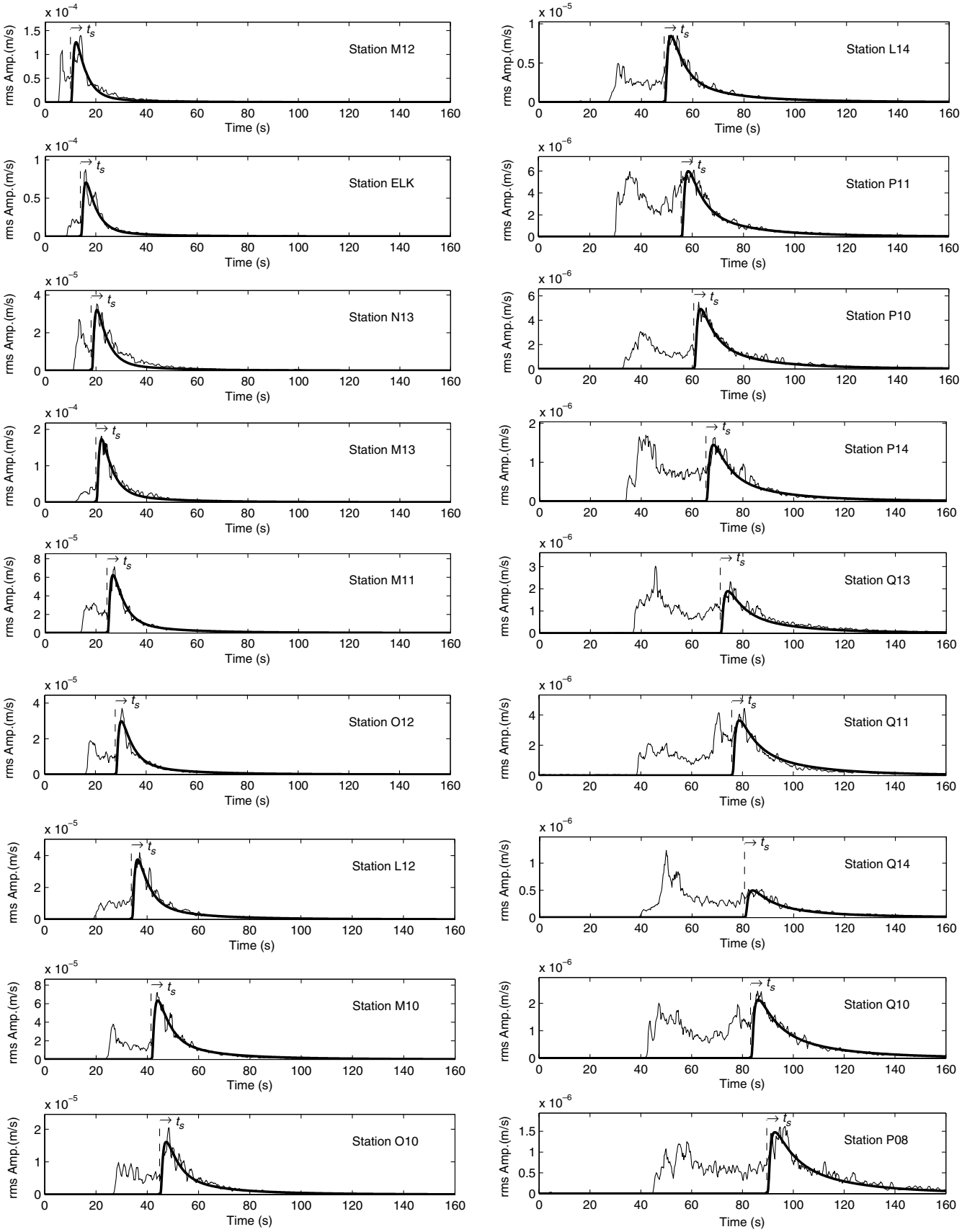


Figure B1. The comparison between the observed rms velocity amplitude (thin line) and synthesized envelopes (thick line).

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